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Clinical Doppelgänger

SMART on FHIR Clinical decision support system

Mia Zhou & Yutong Wu



**Gillings School of
Global Public Health**

Motivation



Problem:

- Traditional dashboards show isolated data such as labs, medications, and diagnoses, but they do not always connect those data points to comparable patient outcomes.

Goal:

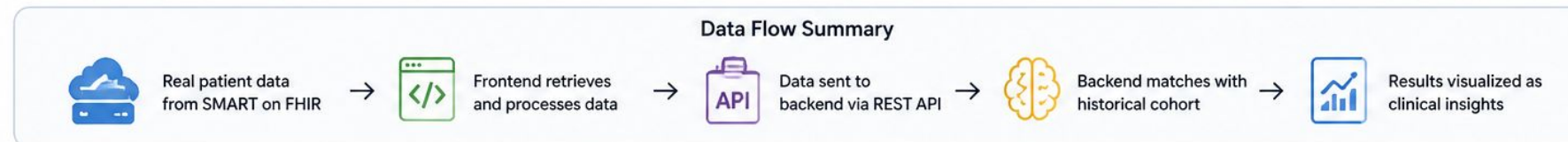
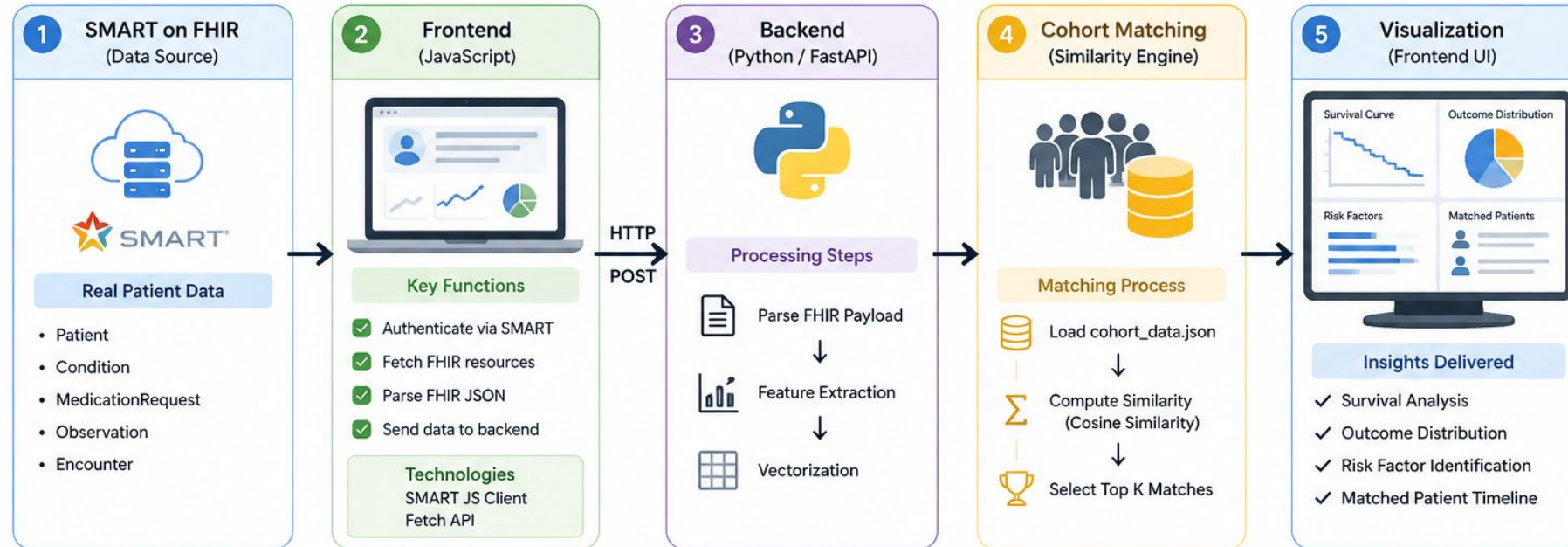
- Find similar patients
- Use historical outcomes to guide decisions

System Overview

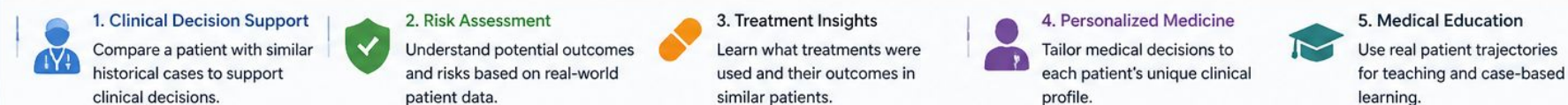


Clinical Doppelganger System Pipeline

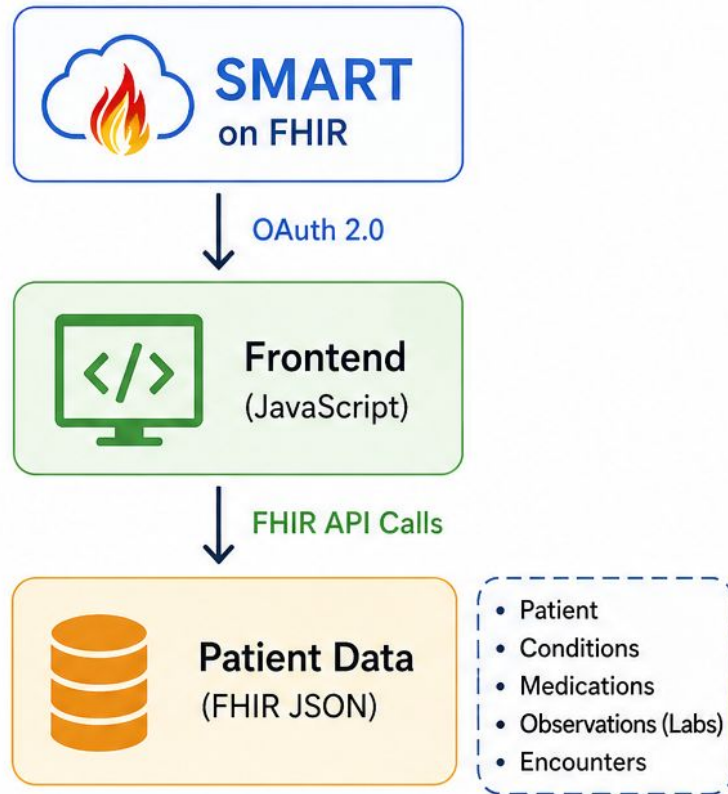
SMART on FHIR + Backend Matching + Visualization



What Can This System Be Used For?



Frontend: Retrieve Real Patient Data with SMART on FHIR



JavaScript Code

```
FHIR.oauth2.ready()
  .then(function(client) {
    var patientId = client.patient.id;
    return Promise.all([
      client.request("Patient/" + patientId),
      client.request(`Condition?patient=${patientId}`),
      client.request(`MedicationRequest?patient=${patientId}`),
      client.request(`Observation?patient=${patientId}`),
      client.request(`Encounter?patient=${patientId}`)
    ]);
  })
  .then(function(responses) {
    const patient = responses[0].entry[0].resource;
    const conditions = responses[1].entry;
    const medications = responses[2].entry;
    const observations = responses[3].entry;
    const encounters = responses[4].entry;
    // Send to backend
    sendToBackend({ patient, conditions, medications,
      observations, encounters });
  });
```

Frontend → Backend



- Convert FHIR data into structured JSON
- Send data via REST API (HTTP POST)
- Backend performs computation and matching
- Results returned to frontend for visualization
-

```
</> JavaScript
fetch("http://localhost:8000/api/fhir-match", {
  method: "POST",
  headers: {
    "Content-Type": "application/json"
  },
  body: JSON.stringify({
    patient: patient,
    conditions: conditions,
    medications: medications,
    observations: observations,
    encounters: encounters,
    topK: 8
  })
})
.then(res => res.json())
.then(data => {
  console.log("AUTO MATCH:", data);
});
```


Backend : Backend: Processing FHIR Data into Features

- Parse raw FHIR JSON into structured data
- Extract clinically meaningful features
- Convert patient into numerical vector
- Enable quantitative comparison between patients

```
def patient_vector(patient):
    conditions = patient.get("conditions") or []
    meds = patient.get("meds") or []

    has_diabetes = any("diabet" in c.lower() for c in conditions)
    has_htn = any("hyper" in c.lower() for c in conditions)

    hba1c = patient.get("latestHbA1c") or 7.5
    bp = patient.get("latestBP") or 130

    return [
        1 if has_diabetes else 0,
        1 if has_htn else 0,
        hba1c / 12,
        bp / 200,
        len(meds),
        len(conditions)
    ]
```

1 Clinical Feature Vector

Representing each patient as a clinical fingerprint

FEATURES USED

Binary (0 or 1)	 Diabetes diagnosis	1
	 Hypertension diagnosis	1
	 Lipid disorder diagnosis	0
	 Metformin use	1
	 ACE inhibitor / ARB use	1
	 Statin use	0
Continuous (Normalized)	 HbA1c value	8.2
	 Systolic blood pressure	145
	 Encounter count	7
	 Medication count	5
	 Diagnosis count	3

NORMALIZATION (Example)

				
HbA1c / 12	Systolic BP / 200	Encounters / 10	Meds Count / 8	Diag Count / 6
8.2	145	7	5	3
÷ 12	÷ 200	÷ 10	÷ 8	÷ 6
= 0.68	= 0.73	= 0.70	= 0.63	= 0.50

Why feature vectors?

- ✓ Convert mixed clinical data into a comparable numeric format.
- ✓ Binary features capture presence of key conditions/medications.
- ✓ Continuous features are normalized so large values do not dominate.

2 Similarity Algorithm

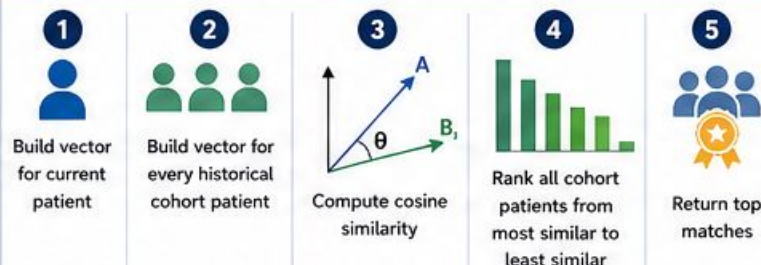
Cosine Similarity – Measuring how close two patients are

COSINE SIMILARITY FORMULA

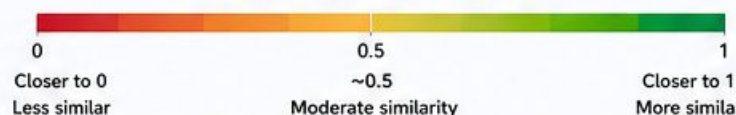
$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

A, B : Vectors
A · B : Dot product
||A|| : Magnitude of A
||B|| : Magnitude of B

MATCHING PROCESS (Step-by-Step)



INTERPRETATION OF SCORES



EXAMPLE CALCULATION

	Current Patient Vector (A)										
	1	1	0	1	1	0	0.68	0.73	0.70	0.63	0.50
	Cohort Patient Vector (B _i)										
	1	0	1	1	1	0	0.58	0.80	0.60	0.50	0.33

Cosine Similarity

$$\text{Similarity}(A, B_i) = \frac{A \cdot B_i}{\|A\| \times \|B_i\|} = 0.81$$

WHY COSINE SIMILARITY?

-  Works well for high-dimensional clinical data with many features.
-  Focuses on pattern similarity rather than raw distance alone.
-  Easy to explain, intuitive, and suitable for clinical presentations.

3 Matched Cohort Data

What is stored in cohort_data.json

WHAT EACH HISTORICAL PATIENT INCLUDES

 ID and Display Name Unique identifier	 Turning Point Key moment or change associated with outcome
 Age and Gender Demographics	 Match Reasons Why this patient was considered similar
 Clinical Features The same features used for matching	 Timeline Important events and measurements over time
 Outcome What happened to the patient	
 Days to Event Number of days from baseline to outcome	

POSSIBLE OUTCOMES

 Improved Patient's condition improved over time with treatment or lifestyle changes.	 Hospitalized Patient experienced a hospitalization related to their condition.	 Deteriorated Patient's condition worsened over time despite treatment or follow-up.
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EXAMPLE MATCH EXPLANATION (Match Reasons)

- ✓ Similar HbA1c range (e.g., 7.5–8.5%)
- ✓ Same medication class (e.g., taking Metformin, ACE inhibitor, Statin)
- ✓ Overlapping diagnoses (e.g., Diabetes, Hypertension)
- ✓ Similar encounter frequency
- ✓ Elevated blood pressure in similar range
- ✓ Care gap or medication gap identified



Speaker Notes

"The matched cases are not just numbers. Each one includes an outcome story, which makes the comparison more interpretable."

Application: What Can This System Be Used For?



1 Clinical Decision Support



- Compare a current patient with similar historical cases
- Provide evidence-based support for clinical decisions

Example:

Find patients like mine and see what happened and what treatments were used.

2 Risk Assessment



- Estimate the risk of future outcomes
- Identify high-risk patients early

Example:

Estimate the risk of hospitalization or poor outcomes for this patient.

3 Treatment Insights



- Learn which treatments worked for similar patients
- Compare outcomes between treatment options

Example:

See which medications or therapies led to better outcomes in similar patients.

4 Personalized Medicine



- Tailor recommendations based on patient similarity
- Support patient-specific care plans

Example:

Provide personalized treatment suggestions based on what worked for similar patients.

5 Medical Education & Research



- Use real patient trajectories for teaching
- Support research with real-world data

Example:

Analyze real cases to teach students or conduct clinical research.



Real-World
Data (FHIR)



Similarity Matching
& Analytics



Actionable
Insights



Better Decisions
Better Outcomes



Our Goal

Empower clinicians with interpretable insights from real-world data to make better decisions and improve patient outcomes.

Demo



Skip

WHAT IS THIS?

A patient comparison workspace

This app helps you start from one patient chart, find past patients who looked clinically similar, and see what happened next. It is a reference tool for chart review, not a prediction engine.

EHR

→

Match

→

Outcome

How do I load a patient? →

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